Youssef Naim

Mechatronics Engineer - Automation and Robotics

Seeking a Permanent Position

+33 6 46 29 45 27 • **contact.naimyoussef@gmail.com** • **https://www.linkedin.com/in/youssef-naim/**

Category B Driving Licence • **Clermont-Ferrand (Mobile in France)** • **Portfolio: https://youssef-naim.github.io/**

Profile

Graduate **Mechatronics** Engineer holding a **Master's Degree (M2) in Automation and Robotics, Mechatronics Track** from **Clermont Auvergne University**. I am seeking a permanent position in the fields of mechatronics, automation, robotics and embedded mechatronic systems. My past experiences have allowed me to acquire key skills that I wish to put to use in technical and innovative projects.

Education

Clermont Auvergne University | 2024 – 2026, Clermont-Ferrand, France
Master's Degree (M2) in Automation and Robotics, Mechatronics Track

Clermont Auvergne University | 2023 – 2024, Clermont-Ferrand, France
Master's Degree (M1) in Automation and Robotics

Clermont Auvergne University | 2022– 2023, Clermont-Ferrand, France
Bachelor's Degree (Licence) in Engineering Sciences, Track: Electronics, Electrical Engineering and Automation (EEA)

Professional Experience and Projects

End-of-Studies Internship: Mechatronics Engineer for the PhénoVigne Application | INRAE, TSCF Unit, Clermont-Ferrand, France | 16 March - 15 September 2026

- **Development** and **validation** of an **active stabilisation** solution for a sensor-carrying mast mounted on a vineyard tractor for phenotyping in viticulture.
- Reconstruction of the PhénoVigne system in **CATIA V5** and extraction of the inertial parameters (***mass***, ***centre of mass***, ***inertia matrix***) required to build the Adams model and for the **dynamic simulation**
- Development of a control strategy in **MATLAB/Simulink** based on the orientation measurements (roll, pitch, yaw) provided by the **IMU** mounted on the mast.
- Setup of a co-simulation **MSC Adams/Simulink** to test and validate the control on a typical test course featuring extreme cross-slope, extreme gradient and abrupt jolts.
- **Validation** of the **control** performance with a tilt error below **1°**, which meets the specification requirements that tolerate an error of **±2°**.
- **Selection** and **sizing of the electrical linear actuators** according to the requirements in terms of **stroke**, **speed** and **force**, as well as the geometric constraints of the system (**space envelope**)

Internship: Assistant Application Engineer in Mobile Robotics | Axter Automation, Vaucresson, France | August - September 2025

- Participation in an **AGV** (Automated Guided Vehicles) application including preparation of electronic components, hardware assembly, programming and configuration, testing, installation and maintenance
- Operation and validation of the proper functioning of the **AGV**

Mapping and Autonomous Navigation with the LIMO Robot | EUPI - UCA, Clermont-Ferrand, France | September 2024 - January 2025

- Implementation of **SLAM** (*Simultaneous Localization and Mapping*) using 2D LiDAR technology to generate an occupancy grid with **ROS 1** and **Python**
- Integration of the **Move_base** package for autonomous navigation combined with **AMCL (Adaptive Monte Carlo Localization)** for real-time localisation of the robot on the map

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Automation of a 24 V Multi-Task fischertechnik Model | EUPI - UCA, Clermont-Ferrand, France | October 2024 - December 2024

- **Design** and **implementation** of a complete automation solution for a multi-process handling station that simulates several processes through which workpieces pass, using a **Siemens Simatic S7-1200** PLC programmed with **TIA Portal**
- Development of an **HMI (Human-Machine Interface)** and execution of functional tests.

End-of-Year Internship: Study and Research Project in Signal Processing | EUPI - UCA, Clermont-Ferrand, France | May 2024 - July 2024

- Acquisition and analysis of real signals in **MATLAB** using the **NI PCIe-6323** module, exploring the impact of sampling frequency, frequency resolution and windowing on signal quality.
- Design of an electronic **PCB** board for educational purposes for a home automation application allowing control of appliances via the simulated **EDF** mains signal (generated by a **GBF** function generator)

Skills

- **Software**: MATLAB, Simulink, CATIA V5, MSC Adams View, LabVIEW, TIA Portal
- **Programming**: Python, C/C++, Linux, Git and GitHub
- **Robotics**: ROS, Modelling, Vendor-specific robot programming (ABB, Staubli), Mobile robotics
- **Automation**: PID control, S7-1200, Networks and IoT (LoRaWAN, MQTT, Modbus, TCP/IP), HMI/SCADA
- **Electronics and Actuators**: Analogue and digital electronics, Microcontrollers (Arduino), EasyEDA, Sensors and Actuators (hydraulic, pneumatic and electric)
- **Soft Skills** : Ability to work in a team, Rigour and organisation, Analytical and synthesis skills

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Languages

- **Arabic**: Native language
- **French**: Bilingual
- **English**: B2 level

Interests and Hobbies

- **Table Tennis (Ping-Pong)**
- **Baby-Foot**
- **Outdoor Hiking and Meditation**